CITI BIKE REBALANCING EXPLORER · DOCUMENTATION SET V2

Document 00 — Index & Reading Order

Where and when does station imbalance become a barrier to riding — and what should NYC DOT do about it?

A four-part written record of a self-directed, public-data-only prototype: the problem it frames, the data and models behind it, the dashboard that presents it, and the process that built it.

Audience: NYC DOT Bike Share & Shared Mobility reviewers, and anyone evaluating the project end-to-end.

Scope: 100% public data (Citi Bike trip archives & GBFS, NYC/NY Open Data, Open-Meteo). $0 hosting stack.

Status: Working prototype. Every number in these documents is reproduced from committed data files by code in `pipeline/`.

Prepared: July 2026. Independent portfolio project — not an official product of NYC DOT or Lyft / Citi Bike.

How to read this set

The four documents are meant to be read in order. Each answers a different question and assumes you have read the ones before it — together they move from *why the project exists* to *how it works*, to *what a user sees*, and finally to *how it was built*.

Document 01 — Problem, Use Case & Solution

The starting point. What imbalance is, why it is a mode-choice barrier rather than only an operations metric, who the intended user is, what data the project stands on, and the three concrete things the tool lets that user do. Read this first: it frames every decision the later documents explain.

Document 02 — Data, Methods, Modeling, EDA, Findings & Limitations

The engine room. The datasets and how they are cleaned; how raw trips become a net-flow signal; the features and the three-tier forecasting model; the station typology, equity join, weather elasticities, and route planner; the honest findings (including where a simple baseline beats the fancy model); the limitations stated plainly; and the roadmap. Read this to trust the numbers.

Document 03 — Dashboard Design, Modes & Metrics

The product. How the analysis is turned into something a reviewer can operate: the two data modes, the map and its color logic, the permanent readout and ranked lists, Investigator Mode's what-if sandbox, and exactly which metrics appear where and what each one means. Read this to understand the interface.

Document 04 — Lessons Learned: AI Collaboration & System-Design Process

The retrospective. How the project was actually built — working with an AI assistant under a fixed set of standing rules, a verify-before-claiming discipline, and a one-module-per-session cadence. Reusable practices for AI-assisted development and for designing a small, honest data system. Read this last: it is the meta-layer over everything above.

> **ONE RULE GOVERNS ALL FOUR DOCUMENTS**
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> Never invent numbers. Every figure quoted here is derived from a committed data file (`data/*.json` , `data/*.parquet`) by code in `pipeline/` , and is reproducible with `python3 -m pipeline.reproduce_all` . Where a number is a *proxy* or a *simplifying assumption* rather than a direct measurement, it is labeled as such in the text, not smoothed over.